ZEEL KHENI

Game Developer, Unity & Unreal Engine

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PROFESSIONAL SUMMARY

Game developer with 6+ years of experience across Unity and Unreal Engine 5, spanning mobile, multiplayer, VR, and WebGL titles. Comfortable owning a feature end to end, from technical design through optimization and release, and equally comfortable working solo or alongside designers, artists, and other engineers. Core strengths include performance optimization and memory management on resource-constrained hardware (iOS and Android), multiplayer networking and lag compensation, and applying SOLID principles and dependency injection (Zenject/Extenject) to keep gameplay architecture clean and maintainable. Additional hands-on experience with audio/rhythm-sync gameplay, live-ops and remote-config tuning, and using AI-assisted tools as part of the day-to-day build process.

CORE TECHNICAL SKILLS

Game Engines: Unity (Expert), Unreal Engine 4/5 (Advanced)

Programming & Architecture: C#, C++, Blueprints, HLSL, JavaScript, Object-Oriented Programming (OOP), SOLID Principles, Design Patterns, Dependency Injection (Zenject/Extenject), Scalable Gameplay Architecture

Mobile Development: iOS and Android builds, mobile performance and memory optimization, cross-platform delivery

Gameplay & Feel: Gameplay Ability System (GAS), Frame Budget & Input Latency Tuning, Physics & Animation Systems, Finite State Machines, Object Pooling, Audio/Rhythm-Sync Gameplay, DSP-Driven Mechanics

Multiplayer & Networking: Photon Fusion, Photon PUN, Mirror, Unreal Dedicated Servers, RPCs, Lag Compensation, Client-Server Prediction, Interpolation

Live Service: Live-Ops, Remote Config, Post-Launch Tuning

Graphics & Rendering: URP, HDRP, Lumen, Nanite, Shader Graph, Niagara, Occlusion Culling, Draw Call Optimization

Profiling & Debugging: Unity Profiler, RenderDoc, Unreal Insights, Asynchronous Programming, Memory Management

DevOps & Workflow: Git, GitHub, GitHub Actions, Jenkins, CI/CD Pipelines, Automated Testing, Agile/Scrum, Code Reviews, AI-Assisted Development Tools

Asset Pipeline: Addressables, Asset Bundles, Content Delivery Optimization

Backend & Cloud: Node.js, MongoDB, Oracle Cloud, PlayFab

Other Platforms: WebGL, VR/XR (XR Interaction Toolkit), Web3 (Moralis, ChainSafe)

PROFESSIONAL EXPERIENCE

Lead Game Engineer & Technical Consultant, *Freelance, Remote* **Dec 2025 - Present**

- Take end-to-end ownership of complex gameplay and multiplayer features across UE5 and Unity client projects, from technical design through implementation, optimization, and release.
- Architect scalable multiplayer systems using Photon Fusion and UE5 dedicated servers, with lag compensation and client-server state disagreement as the core recurring technical challenge.
- Apply SOLID principles and dependency injection patterns to build modular, maintainable gameplay architecture for digital twin and metaverse client projects, including MetaHuman integration for runtime believability.
- Delivered Snap Orbit (Unity WebGL) for YouTube Playables, engineering aggressive asset streaming to hit sub-5-second load times on mobile browsers.
- Extend delivery to iOS builds alongside Android, optimizing performance and memory usage for Apple hardware as part of cross-platform mobile delivery.
- Use AI-assisted tools as part of the regular build and iteration workflow, from prototyping through debugging.
- Own game server infrastructure (Node.js, MongoDB, Oracle Cloud/Ubuntu), balancing uptime with cost efficiency.
- Built custom Unity and UE5 editor tools that measurably reduced artist time spent on asset ingestion and lighting bakes.

Senior Unity Developer, *Noborder.z Innovations, Indore, India* **Jun 2024 - Dec 2025**

- Led migration of core metaverse platform architecture from Unity to Unreal Engine 5, leveraging Nanite and Lumen to significantly improve visual fidelity.
- Designed and implemented multiplayer logic for amusement park ride systems in UE5 (C++, Blueprints), solving multi-client physics synchronization through iterative architectural refinement.

- Diagnosed and resolved slow WebGL first-load performance by restructuring Addressables and deferring upfront asset loading, cutting load times by 30%.
- Built VR systems using the XR Interaction Toolkit, profiling and optimizing the render pipeline to hold performance across varied hardware.

Senior Unity Developer, *Technomancy Private Limited, Indore, India* **Oct 2022 - Jun 2024**

- Architected the multiplayer backbone for Stardust Metaverse on Photon, engineering persistent session state management under continuous player join/leave activity, a live-ops-style stability problem in practice.
- Resolved recurring desync issues by reworking RPC batching and interpolation logic, measurably improving stability and reducing user complaints.
- Integrated Web3 functionality (Moralis, ChainSafe): wallet authentication, NFT rendering, contract event listeners, closing the gap between demo and production reliability.
- Developed custom URP shaders and lighting pipelines, maintaining performance through continuous profiling.
- Implemented GitHub Actions and Jenkins CI/CD pipelines, automating builds and reducing manual sprint overhead.

Senior Unity Developer, *InfoOyster Technologies, Bhavnagar, India* **Jun 2022 - Oct 2022**

- Built a hybrid Photon/Mirror multiplayer pipeline, reducing sync conflicts by approximately 30%.
- Implemented blockchain wallet login and smart contract support for a decentralized game client.
- Used Unity Profiler and RenderDoc to isolate and resolve CPU spikes, converting vague frame-timing complaints into fixable, documented issues.

Unity Developer, *InfoOyster Technologies, Bhavnagar, India* **Mar 2021 - Jun 2022**

- Rebuilt the character controller and customization system using object pooling, improving runtime performance by approximately 50%.
- Reduced memory usage by 20% through deliberate scene and asset lifecycle management.
- Rewrote animation logic with finite state machines (FSMs) and Animator state machines, simplifying multiplayer edge-case handling through predictable state transitions.

Unity Developer, *Freelance, Surat, India* **May 2020 - Mar 2021**

- Independently built and shipped Android games end-to-end (UI, animation, gameplay), maintaining a 99% crash-free rate across releases.
- Built and optimized 3D environments in ProBuilder for mobile GPU constraints, prioritizing draw-call reduction over polygon count.

KEY PROJECTS

- XANA Metaverse: Photon Fusion multiplayer, synchronized ride physics, scene optimization and LOD work for a live production platform.
- Run For Money: Server-authoritative architecture with PlayFab backend, built for reliable competitive play.
- Stardust Metaverse: Persistent Photon multiplayer world with Web3 wallet login and NFT integration.
- Dikanz / Vaikunth Metaverse: NFT display, wallet authentication, multiplayer customizable spaces, shipped to production users.
- Wild Waker, HexGlow, IQ-Ball, Difference Journey, Haste Runner: Puzzle, action, and physics titles across genres, full loops built solo.

EDUCATION

Bachelor of Engineering, Information Technology, Gujarat Technological University, 2022
Diploma, Information Technology, Gujarat Technological University, 2019